6 Tabelle spezieller Laplace-Transformationen

	Bildfunktion $F(s)$	Originalfunktion $f(t)$
(1)	1	$\delta(t)$ (*Diracsche* Deltafunktion)
(2)	$\frac{1}{s}$	$\sigma(t)$ (Sprungfunktion, Sigmafunktion)
(3)	$\frac{1}{s-a}$	e^{at}
(4)	$\frac{1}{s^2}$	t
(5)	$\frac{1}{s(s-a)}$	$\frac{e^{at}-1}{a}$
(6)	$\frac{1}{(s-a)^2}$	$t \cdot e^{at}$
(7)	$\frac{1}{(s-a)(s-b)}$	$\frac{e^{at}-e^{bt}}{a-b}$
(8)	$\frac{s}{(s-a)^2}$	$(1+at) \cdot e^{at}$
(9)	$\frac{s}{(s-a)(s-b)}$	$\frac{a \cdot e^{at} - b \cdot e^{bt}}{a-b}$
(10)	$\frac{1}{s^3}$	$\frac{1}{2} t^2$
(11)	$\frac{1}{s^2(s-a)}$	$\frac{e^{at}-at-1}{a^2}$
(12)	$\frac{1}{s(s-a)^2}$	$\frac{(at-1) \cdot e^{at}+1}{a^2}$
(13)	$\frac{1}{(s-a)^3}$	$\frac{1}{2} t^2 \cdot e^{at}$
(14)	$\frac{1}{s(s-a)(s-b)}$	$\frac{b \cdot e^{at} - a \cdot e^{bt} + a - b}{ab(a-b)}$
(15)	$\frac{1}{(s-a)(s-b)(s-c)}$	$\frac{(b-c) \cdot e^{at} + (c-a) \cdot e^{bt} + (a-b) \cdot e^{ct}}{(a-b)(a-c)(b-c)}$
(16)	$\frac{s}{(s-a)^3}$	$\left(\frac{1}{2} at^2 + t\right) \cdot e^{at}$
(17)	$\frac{s}{(s-a)(s-b)^2}$	$\frac{a \cdot e^{at} - [a + b(a-b)t] \cdot e^{bt}}{(a-b)^2}$
(18)	$\frac{s}{(s-a)(s-b)(s-c)}$	$\frac{a(b-c) \cdot e^{at} + b(c-a) \cdot e^{bt} + c(a-b) \cdot e^{ct}}{(a-b)(a-c)(b-c)}$

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(19)	$\frac{s^2}{(s-a)^3}$	$\left(\frac{1}{2}a^2t^2+2at+1\right)\cdot e^{at}$
(20)	$\frac{s^2}{(s-a)(s-b)^2}$	$\frac{a^2\cdot e^{at}-[b^2(a-b)t+2ab-b^2]\cdot e^{bt}}{(a-b)^2}$
(21)	$\frac{s^2}{(s-a)(s-b)(s-c)}$	$\frac{a^2(b-c)\cdot e^{at}+b^2(c-a)\cdot e^{bt}+c^2(a-b)\cdot e^{ct}}{(a-b)(a-c)(b-c)}$
(22)	$\frac{1}{s^n}\quad(n\in\mathbb{N}^*)$	$\frac{t^{n-1}}{(n-1)!}$
(23)	$\frac{1}{(s-a)^n}\quad(n\in\mathbb{N}^*)$	$\frac{t^{n-1}\cdot e^{at}}{(n-1)!}$
(24)	$\frac{1}{s^2+a^2}$	$\frac{\sin(at)}{a}$
(25)	$\frac{s}{s^2+a^2}$	$\cos(at)$
(26)	$\frac{(\sin b)\cdot s+a\cdot\cos b}{s^2+a^2}$	$\sin(at+b)$
(27)	$\frac{(\cos b)\cdot s-a\cdot\sin b}{s^2+a^2}$	$\cos(at+b)$
(28)	$\frac{1}{(s-b)^2+a^2}$	$\frac{e^{bt}\cdot\sin(at)}{a}$
(29)	$\frac{s-b}{(s-b)^2+a^2}$	$e^{bt}\cdot\cos(at)$
(30)	$\frac{1}{s^2-a^2}$	$\frac{\sinh(at)}{a}$
(31)	$\frac{s}{s^2-a^2}$	$\cosh(at)$
(32)	$\frac{1}{(s-b)^2-a^2}$	$\frac{e^{bt}\cdot\sinh(at)}{a}$
(33)	$\frac{s-b}{(s-b)^2-a^2}$	$e^{bt}\cdot\cosh(at)$
(34)	$\frac{1}{s(s^2+4a^2)}$	$\frac{\sin^2(at)}{2a^2}$
(35)	$\frac{s^2+2a^2}{s(s^2+4a^2)}$	$\cos^2(at)$
(36)	$\frac{1}{s(s^2+a^2)}$	$\frac{1-\cos(at)}{a^2}$

	Bildfunktion $F(s)$	Originalfunktion $f(t)$
(37)	$\frac{1}{(s^2+a^2)^2}$	$\frac{\sin(at) - at \cdot \cos(at)}{2a^3}$
(38)	$\frac{s}{(s^2+a^2)^2}$	$\frac{t \cdot \sin(at)}{2a}$
(39)	$\frac{s^2}{(s^2+a^2)^2}$	$\frac{\sin(at) + at \cdot \cos(at)}{2a}$
(40)	$\frac{s^2-a^2}{(s^2+a^2)^2}$	$t \cdot \cos(at)$
(41)	$\frac{s^3}{(s^2+a^2)^2}$	$\cos(at) - \frac{1}{2}\, at \cdot \sin(at)$
(42)	$\frac{1}{s^2(s^2+a^2)}$	$\frac{at - \sin(at)}{a^3}$
(43)	$\frac{1}{(s^2+a^2)(s^2+b^2)}$	$\frac{a \cdot \sin(bt) - b \cdot \sin(at)}{ab(a^2-b^2)}$
(44)	$\frac{s}{(s^2+a^2)(s^2+b^2)}$	$\frac{\cos(bt) - \cos(at)}{a^2-b^2}$
(45)	$\frac{s^2}{(s^2+a^2)(s^2+b^2)}$	$\frac{a \cdot \sin(at) - b \cdot \sin(bt)}{a^2-b^2}$
(46)	$\frac{s^3}{(s^2+a^2)(s^2+b^2)}$	$\frac{a^2 \cdot \cos(at) - b^2 \cdot \cos(bt)}{a^2-b^2}$
(47)	$\frac{1}{s(s^2+a^2)^2}$	$\frac{-at \cdot \sin(at) - \cos(at) + 1}{2a^4}$
(48)	$\frac{1}{s(s^2+a^2)(s^2+b^2)}$	$\frac{b^2 \cdot \cos(at) - a^2 \cdot \cos(bt) + a^2 - b^2}{a^2b^2(a^2-b^2)}$
(49)	$\frac{1}{s(s^2-a^2)}$	$\frac{\cosh(at) - 1}{a^2}$
(50)	$\frac{1}{(s^2-a^2)^2}$	$\frac{at \cdot \cosh(at) - \sinh(at)}{2a^3}$
(51)	$\frac{s}{(s^2-a^2)^2}$	$\frac{t \cdot \sinh(at)}{2a}$
(52)	$\frac{s^2}{(s^2-a^2)^2}$	$\frac{\sinh(at) + at \cdot \cosh(at)}{2a}$
(53)	$\frac{s^2+a^2}{(s^2-a^2)^2}$	$t \cdot \cosh(at)$
(54)	$\frac{s^3}{(s^2-a^2)^2}$	$\frac{1}{2}\, at \cdot \sinh(at) + \cosh(at)$

	Bildfunktion $F(s)$	Originalfunktion $f(t)$
(55)	$\frac{1}{s^2(s^2-a^2)}$	$\frac{\sinh(at)-at}{a^3}$
(56)	$\frac{1}{s(s^2-a^2)^2}$	$\frac{at\cdot\sinh(at)-2\cdot\cosh(at)+2}{a^4}$
(57)	$\frac{1}{s^3+a^3}$	$\frac{1}{3a^2}\left[\sqrt{3}\cdot\sin\left(\frac{\sqrt{3}at}{2}\right)-\cos\left(\frac{\sqrt{3}at}{2}\right)+e^{-3at/2}\right]\cdot e^{at/2}$
(58)	$\frac{s}{s^3+a^3}$	$\frac{1}{3a}\left[\sqrt{3}\cdot\sin\left(\frac{\sqrt{3}at}{2}\right)+\cos\left(\frac{\sqrt{3}at}{2}\right)-e^{-3at/2}\right]\cdot e^{at/2}$
(59)	$\frac{s^2}{s^3+a^3}$	$\frac{1}{3}\left[e^{-at}+2\cdot e^{at/2}\cdot\cos\left(\frac{\sqrt{3}at}{2}\right)\right]$
(60)	$\frac{1}{s^3-a^3}$	$\frac{1}{3a^2}\left[e^{3at/2}-\sqrt{3}\cdot\sin\left(\frac{\sqrt{3}at}{2}\right)-\cos\left(\frac{\sqrt{3}at}{2}\right)\right]\cdot e^{-at/2}$
(61)	$\frac{s}{s^3-a^3}$	$\frac{1}{3a}\left[e^{3at/2}+\sqrt{3}\cdot\sin\left(\frac{\sqrt{3}at}{2}\right)-\cos\left(\frac{\sqrt{3}at}{2}\right)\right]\cdot e^{-at/2}$
(62)	$\frac{s^2}{s^3-a^3}$	$\frac{1}{3}\left[e^{at}-2\cdot e^{-at/2}\cdot\cos\left(\frac{\sqrt{3}at}{2}\right)\right]$
(63)	$\frac{1}{s^4+a^4}$	$\frac{1}{a^3\sqrt{2}}\left[\sin\left(\frac{at}{\sqrt{2}}\right)\cdot\cosh\left(\frac{at}{\sqrt{2}}\right)-\cos\left(\frac{at}{\sqrt{2}}\right)\cdot\sinh\left(\frac{at}{\sqrt{2}}\right)\right]$
(64)	$\frac{s}{s^4+a^4}$	$\frac{\sin\left(\frac{at}{\sqrt{2}}\right)\cdot\sinh\left(\frac{at}{\sqrt{2}}\right)}{a^2}$
(65)	$\frac{s^2}{s^4+a^4}$	$\frac{1}{a\sqrt{2}}\left[\cos\left(\frac{at}{\sqrt{2}}\right)\cdot\sinh\left(\frac{at}{\sqrt{2}}\right)+\sin\left(\frac{at}{\sqrt{2}}\right)\cdot\cosh\left(\frac{at}{\sqrt{2}}\right)\right]$
(66)	$\frac{s^3}{s^4+a^4}$	$\cos\left(\frac{at}{\sqrt{2}}\right)\cdot\cosh\left(\frac{at}{\sqrt{2}}\right)$
(67)	$\frac{1}{s^4-a^4}$	$\frac{\sinh(at)-\sin(at)}{2a^3}$
(68)	$\frac{s}{s^4-a^4}$	$\frac{\cosh(at)-\cos(at)}{2a^2}$
(69)	$\frac{s^2}{s^4-a^4}$	$\frac{\sinh(at)+\sin(at)}{2a}$
(70)	$\frac{s^3}{s^4-a^4}$	$\frac{\cosh(at)+\cos(at)}{2}$
(71)	$\frac{s}{s^4+4a^4}$	$\frac{\sin(at)\cdot\sinh(at)}{2a^2}$

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(72)	$\dfrac{s^2 - 2a^2}{s^4 + 4a^4}$	$\dfrac{\cos(at) \cdot \sinh(at)}{a}$
(73)	$\dfrac{s^2 + 2a^2}{s^4 + 4a^4}$	$\dfrac{\sin(at) \cdot \cosh(at)}{a}$
(74)	$\dfrac{s^3}{s^4 + 4a^4}$	$\cos(at) \cdot \cosh(at)$
(75)	$\dfrac{1}{\sqrt{s}}$	$\dfrac{1}{\sqrt{\pi t}}$
(76)	$\dfrac{1}{s\sqrt{s}}$	$2 \cdot \sqrt{\dfrac{t}{\pi}}$
(77)	$\dfrac{1}{s^2\sqrt{s}}$	$\dfrac{4}{3}t \cdot \sqrt{\dfrac{t}{\pi}}$
(78)	$\dfrac{s + a}{s\sqrt{s}}$	$\dfrac{1 + 2at}{\sqrt{\pi t}}$
(79)	$\dfrac{1}{\sqrt{s + a}}$	$\dfrac{e^{-at}}{\sqrt{\pi t}}$
(80)	$\sqrt{s - a} - \sqrt{s - b}$	$\dfrac{e^{bt} - e^{at}}{2t \cdot \sqrt{\pi t}}$
(81)	$\dfrac{1}{s^n\sqrt{s}} \quad (n \in \mathbb{N}^*)$	$\dfrac{4^n \cdot n! \cdot t^{(2n-1)/2}}{(2n)!\sqrt{\pi}}$
(82)	$\ln\left(\dfrac{s - a}{s}\right)$	$\dfrac{1 - e^{at}}{t}$
(83)	$\ln\left(\dfrac{s - a}{s - b}\right)$	$\dfrac{e^{bt} - e^{at}}{t}$
(84)	$\ln\left(\dfrac{s + a}{s - a}\right)$	$\dfrac{2 \cdot \sinh(at)}{t}$
(85)	$\ln\left(\dfrac{s^2 + a^2}{s^2}\right)$	$\dfrac{2[1 - \cos(at)]}{t}$
(86)	$\ln\left(\dfrac{s^2 + a^2}{s^2 + b^2}\right)$	$\dfrac{2[\cos(bt) - \cos(at)]}{t}$
(87)	$\arctan\left(\dfrac{a}{s}\right)$	$\dfrac{\sin(at)}{t}$
(88)	$\arctan\left(\dfrac{2as}{s^2 - a^2 + b^2}\right)$	$\dfrac{2 \cdot \sin(at) \cdot \cos(bt)}{t}$
(89)	$\arctan\left(\dfrac{s^2 - as + b}{ab}\right)$	$\dfrac{(e^{at} - 1) \cdot \sin(bt)}{t}$